A GORENSTEIN PROPER CURVE OF GENUS ZERO IS A PLANE CONIC

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the degree computation, the closed immersion, the genus restriction to d in {1, 2}, and the linear-independence exclusion are all retained verbatim.

Editorial note. The label lemma-genus-zero names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Degree of the dualizing sheaf in genus zero. Same chapter, Lemma lemma-genus-gorenstein. Gives deg(omega_X) = -2, hence deg(omega_X^tensor -1) = 2.
- Closed immersion from a degree-2 invertible sheaf. Same chapter, Lemma lemma-genus-zero-positive-degree. Turns a basis of the three-dimensional space of global sections of L = omega_X^tensor -1 into a closed immersion into the projective plane.
- Plane-curve arithmetic genus. Same chapter, Lemmas lemma-equation-plane-curve and lemma-genus-plane-curve. Writes the equation F of the plane curve of degree d and forces d in {1, 2} from the vanishing genus.

Source reference. Original reference 0BS2: sources/stacks-curves/pinned-original-source.tex, source lines 796–796. Prerequisite pointer invoked at proof line(s) 2116; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 0BYB: sources/stacks-curves/pinned-original-source.tex, source lines 1859–1859. Prerequisite pointer invoked at proof line(s) 2127; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C19: sources/stacks-curves/pinned-original-source.tex, source lines 1744–1744. Prerequisite pointer invoked at proof line(s) 2117; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0BYD: sources/stacks-curves/pinned-original-source.tex, source lines 1904–1904. Prerequisite pointer invoked at proof line(s) 2129; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C6T: sources/stacks-curves/pinned-original-source.tex, source lines 2019–2019. Prerequisite pointer invoked at proof line(s) 2119; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a proper curve over a field k with $H^0(X, \mathcal{O}_X) = k$. If X is Gorenstein and has genus 0, then X is isomorphic to a plane curve of degree 2.*

Proof. Consider the invertible sheaf $\mathcal{L} = \omega_X^{\otimes -1}$ where ω_X is as in Lemma 0BS2. Then $\deg(\omega_X) = -2$ by Lemma 0C19 and hence $\deg(\mathcal{L}) = 2$. By Lemma 0C6T we conclude that choosing a basis s_0, s_1, s_2 of the k-vector space of global sections of $\mathcal{L}$ we obtain a closed immersion

$$\varphi_{(\mathcal{L},(s_0,s_1,s_2))} : X \longrightarrow \mathbf{P}^2_k$$

Thus X is a plane curve of some degree d. Let $F \in k[T_0, T_1, T_2]_d$ be its equation (Lemma 0BYB). Because the genus of X is 0 we see that d is 1 or 2 (Lemma 0BYD). Observe that F restricts to the zero section on $\varphi(X)$ and hence $F(s_0, s_1, s_2)$ is the zero section of $\mathcal{L}^{\otimes 2}$. Because s_0, s_1, s_2 are linearly independent we see that F cannot be linear, i.e., $d = \deg(F) \geq 2$. Thus $d = 2$ and the proof is complete. $\square$